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Peak.News

Where satellite data silences climate denial



Institution

UEES - Ecuador

Track

Climate Change and the Alpine Information Battle



Introduction

The Reason

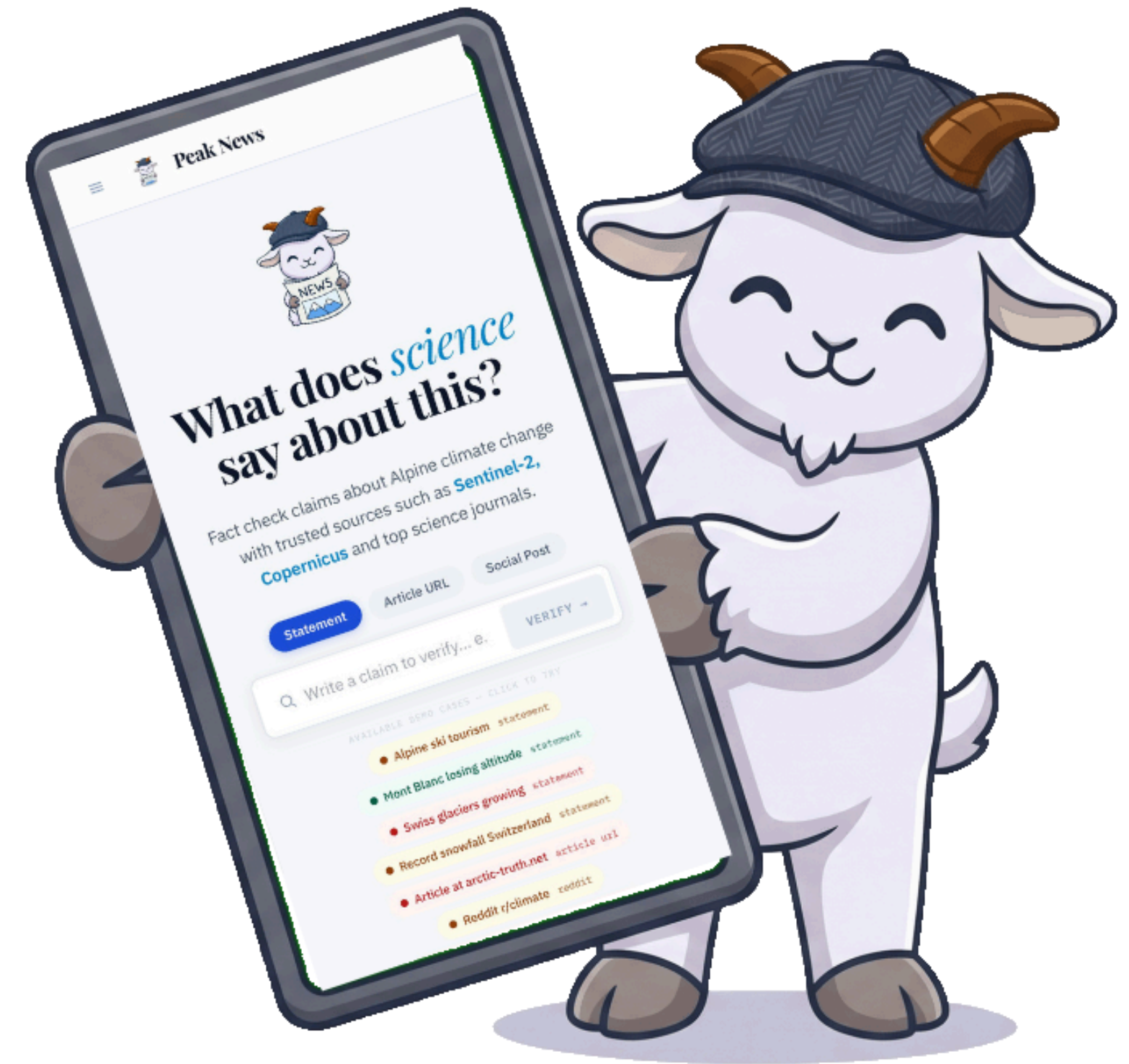
Despite abundant scientific evidence, the Alpine region is under threat from the 'New Denial' sophisticated disinformation campaigns that minimize glacier retreat and delay climate action. Journalists lack the technical tools to rapidly verify these deceptive claims against complex Earth Observation data. PeakNews was created to bridge this gap.



[ABOUT US](#)[MAPS](#)[ANALYSIS](#)[UEES-ECU](#)

What is Peak News?

PEAKNEWS is a web-based verification platform that empowers users to cross-reference suspicious news with high-resolution satellite imagery and peer-reviewed science. The platform provides a transparent "veracity score" that moves the climate conversation from speculative headlines to objective, geographic proof.





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MAPS

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Peak News
ALPINE FACT CHECKER

VERIFYSOURCESMETHODOLOGY

LIVEALPES

What does *science* say about this?

Fact check claims about Alpine climate change with trusted sources such as Sentinel-2, Copernicus and top science journals.

StatementArticle URLSocial Post

VERIFY

AVAILABLE SOME CLAIMS - CLICK TO TRY

Alpine ski tourismstatement

Mont Blanc losing altitudestatement

Swiss glaciers growingstatement

Record snowfall Switzerlandstatement

Article at arctic-truth.netarticle url

Reddit r/climatereddit

Tweet @Kursed65936965x/twitter

131 VERIFIED

46 FALSE DETECTED

94% ACCURACY

SCIENTIFIC REFERENCE REPOSITORIES

NatureIPCCWHO CopernicusScienceAGUGLAMOS

Peak News
ALPINE FACT CHECKER

VERIFYSOURCESMETHODOLOGY

LIVEALPES

StatementArticle URLSocial Post

VERIFY

ANALYSED CLAIM

"Swiss glaciers are growing in 2025"

MISLEADING52/100

Analysed: 28 March 2026

Misleading claim: Alpine ski tourism has not disappeared, but is threatened

-41.1%
Snow coverage Tyrol
2000-2025 - GEE / Landsat 8

-62%
Pasterze glacier retreat
2000-2025 - GLAMOS / Sentinel-2

5,937 ha
Historic minimum 2022
low at 2009 baseline - record low

>1,800 m
Operational altitude
Swiss Alps with natural snow decline - 400

ANALYSIS

Snow coverage data from Tyrol (Austria) processed with Google Earth Engine shows an unambiguous trend: from 15,603 hectares in 2000 to 9,188 hectares in 2025, a net decline of

SOURCES USED - 6 REPOSITORIES - PEER-REVIEWED LITERATURE

SATELLITE DATA & MONITORING

1 Sentinel-2 (ESA)
OPTICAL SATELLITE - 10M
sentinel.esa.int97%

2 Landsat 8/9 (USGS)
MULTISPECTRAL SATELLITE - 30M
landsat.usgs.gov95%

3 GLAMOS Switzerland
GLACIER MONITORING NETWORK
glamos.ch99%

4 MeteoSwiss
METEOROLOGICAL DATA
meteoswiss.admin.ch96%

5 Semantic Scholar
SCIENTIFIC LITERATURE
semantic scholar.org88%

6 IPCC AR6
PEER-REVIEWED SCIENTIFIC REPORT
ipcc.ch100%

SCIENTIFIC REFERENCE REPOSITORIES

Nature Climate ChangeNatuze

IPCC Sixth Assessment ReportIPCC

ALPINE FACT CHECKER

StatementArticle URLSocial Post

VERIFY

ANALYSED CLAIM

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SATELLITE INDICATORS

Glacier retreat
since 200062%

Snow coverage
current year61%

Trend
vs. 1970-29.5%

NDVI Index
vegetation 2 Jan0.41

DETECTED KEYWORDS

no longerimpossibledisappearedAlps

glaciersski

Peak News
ALPINE FACT CHECKER

VERIFYSOURCESMETHODOLOGY

LIVEALPES

StatementArticle URLSocial Post

VERIFY

GEE CHARTS - STATISTICAL ANALYSIS

Charts generated with Google Earth Engine from Landsat 8 and Sentinel-2 data. Scroll to see the full analysis.

2. Cumulative loss vs. final - Total, years, hectares

3. Proportion per decade - Total, years

1 Annual snow coverage trend (hectares). The regression line confirms a statistically significant multi-decade decline across the study region.

2 Net change: baseline year vs. latest year. Bars show the initial and final snow volume — the percentage figure is derived from Landsat 8 pixel counts.

3 Decade-by-decade average. Each bar groups a full ten-year period, making the acceleration of decline between 2000 and 2020 clearly visible.

4 Anomaly vs. historical baseline. Years above/below the long-term mean are highlighted, showing how frequently recent years fall into extreme-deficit territory.

5 Year-on-year variability. Short-term fluctuations (including anomalously snowy years) are shown against the structural downward trend — confirming that record snowfalls do not

SOURCES USED - 6 REPOSITORIES - PEER-REVIEWED LITERATURE

SATELLITE INDICATORS

Glacier retreat
since 200062%

Snow coverage
current year61%

Trend
vs. 1970-29.5%

NDVI Index
vegetation 2 Jan0.41

DETECTED KEYWORDS

no longerimpossibledisappearedAlps

glaciersski

TRUTHFULNESS SCALE

Verified

Mostly true52/100

Misleading

False



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About Us



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Juan Sotomayor



Karel Gonzalez



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Paula Martillo

UEES - 3RD YEAR
Computer Science
Engineer

Strategist in
creating high-fidelity
technological
implementations.

Expert in
transforming
abstract scientific
data into intuitive
graphical interfaces.



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Karel Gonzalez



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Justin Soledispa

UEES - 3RD YEAR
Computer Science
Engineer

Specialist in
multivariate analysis
and predictive models
for the automated
detection of
informational bias.

Experienced in
designing data-driven
decision systems and
explainable AI
workflows.



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Juan Sotomayor

UEES - 3RD YEAR
Computer Science
Engineer

Specialist in Linux
operating systems,
hardware optimization,
and the maintenance
of critical
infrastructures.

Ensures the stability
and performance of
intensive processing
systems.



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Justin Soledispa



Juan Sotomayor



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UEES - 3RD YEAR
Computer Science
Engineer

Expert in optimizing
spatial queries and
managing data
integrity across large
volumes of historical
data.

Specialist in data
modeling, scalable
storage solutions, and
high-performance
analytical
environments.



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Daniel Vaca

UEES - 3RD YEAR
Computer Science
Engineer

Technical strategist
responsible for
system modularity and
the integration of high-
availability cloud
services.

Specialist in
distributed systems,
backend
orchestration, and
resilient platform
engineering.

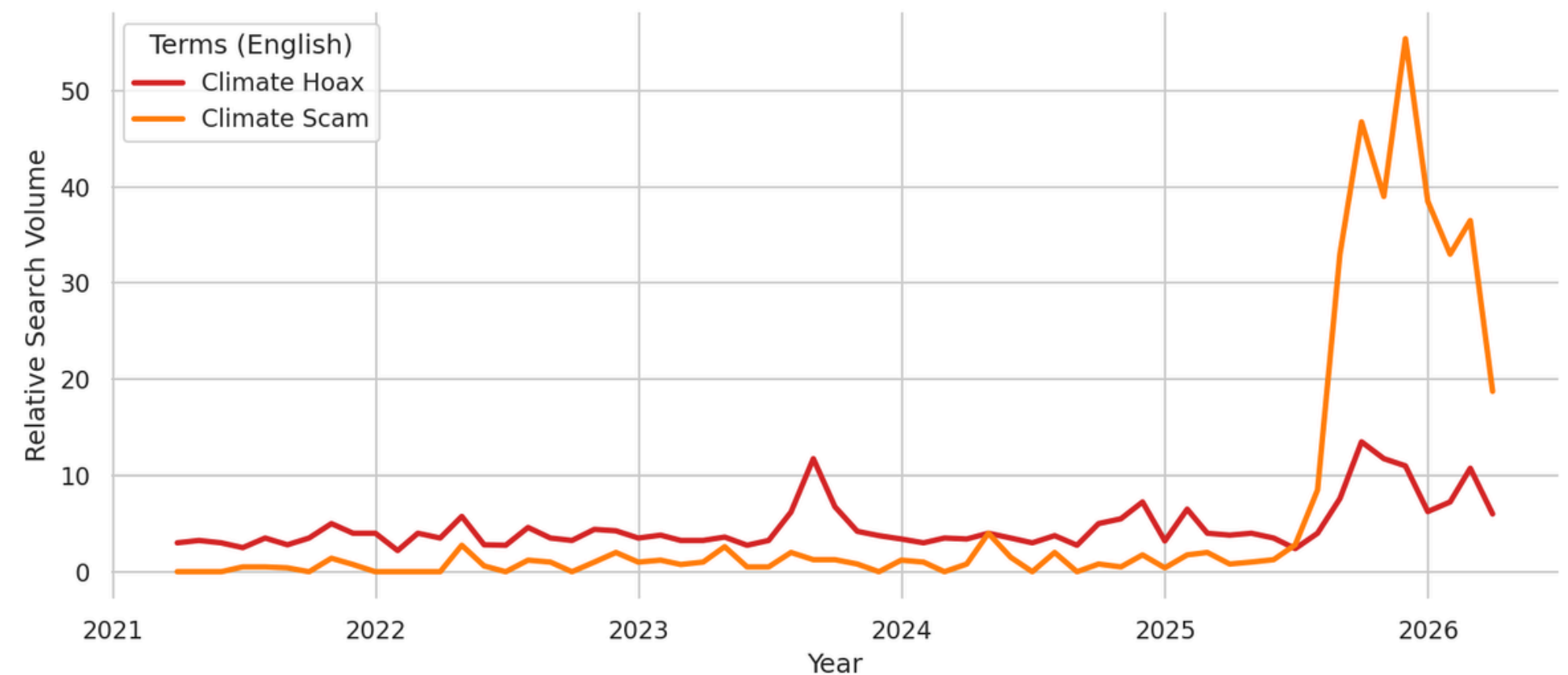


How Is Misinformation A Problem?

Misinformation has become a major problem today because we are constantly exposed to unverified content through social media and news feeds. Platforms like X (formerly Twitter), Facebook, and TikTok amplify it, with over 72% of global internet users encountering false info monthly and 52% weekly on Facebook alone. This rapid spread—fake news travels 10x faster than truth—erodes trust and delays action on issues like climate change.

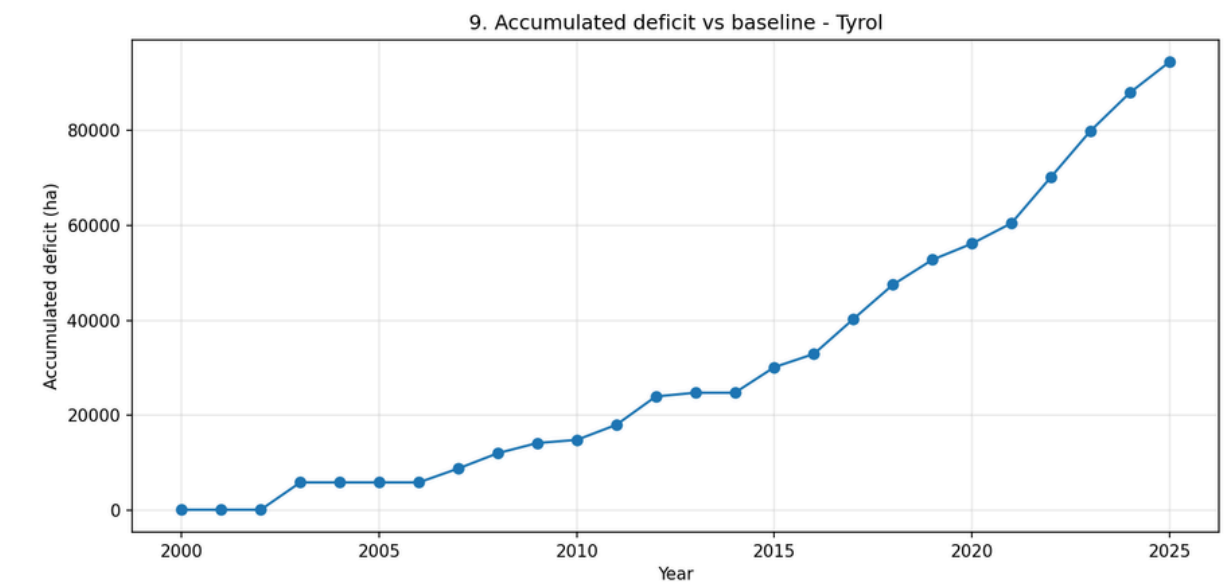
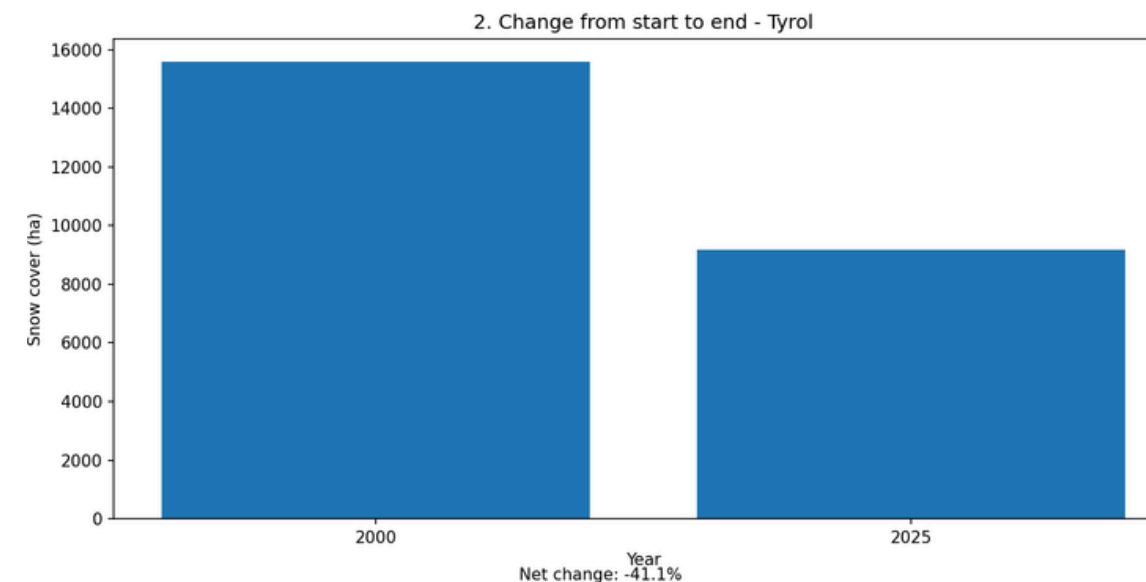
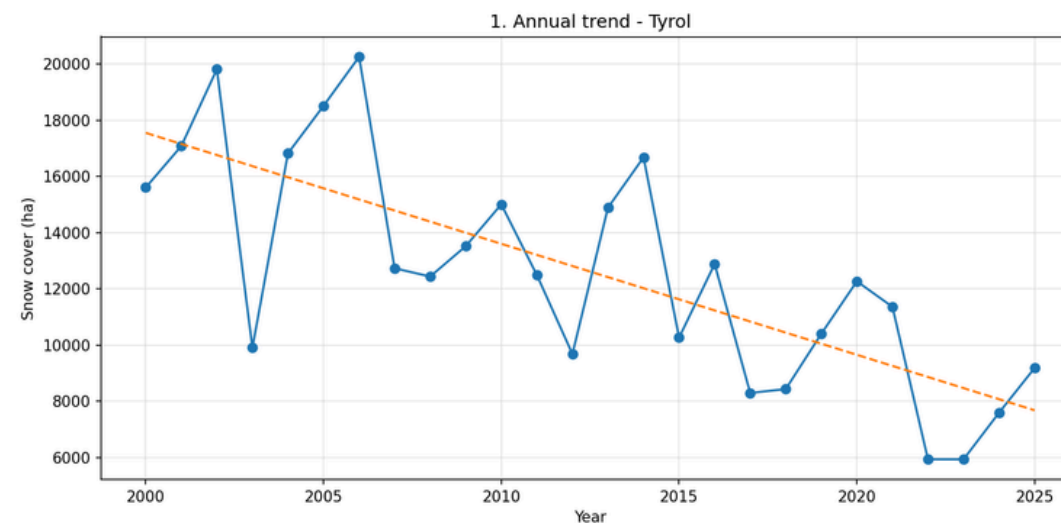
PEAK NEWS combats climate misinformation on Alpine issues by delivering AI-powered, satellite-verified fact-checks for journalists in seconds. It ingests news claims, cross-references Earth observation data and scientific papers, then outputs a rigorous traffic-light verdict: Verified (satellite-backed), Unverifiable (data gaps), or False (contradicts evidence).

Increase in Climate Disinformation Searches (Last 5 Years)





Data We Found - Snow Coverage - Austria



Annual snow-covered area in Tirol (Ötztal Alps), Austria, 2000–2025.

Derived from MODIS/Terra MOD10A1.061 Snow Cover Daily Global 500 m in Google Earth Engine. The area of interest was defined around 46.750°N, 10.917°E with a 22 km buffer. Annual values represent the estimated snow-covered area (ha) from August composites, after QA filtering and snow classification using NDSI > 40.

Source: NASA NSIDC MODIS/Terra MOD10A1.061 (500 m, daily); processed in Google Earth Engine.

Change in snow-covered area between 2000 and 2025 in Tirol (Ötztal Alps), Austria.

Based on annual snow-cover estimates derived from MODIS/Terra MOD10A1.061. The comparison highlights the net percentage change in August snow-covered area across the study period.

Source: NASA NSIDC MODIS/Terra MOD10A1.061 (500 m, daily); analysis by authors in Google Earth Engine.

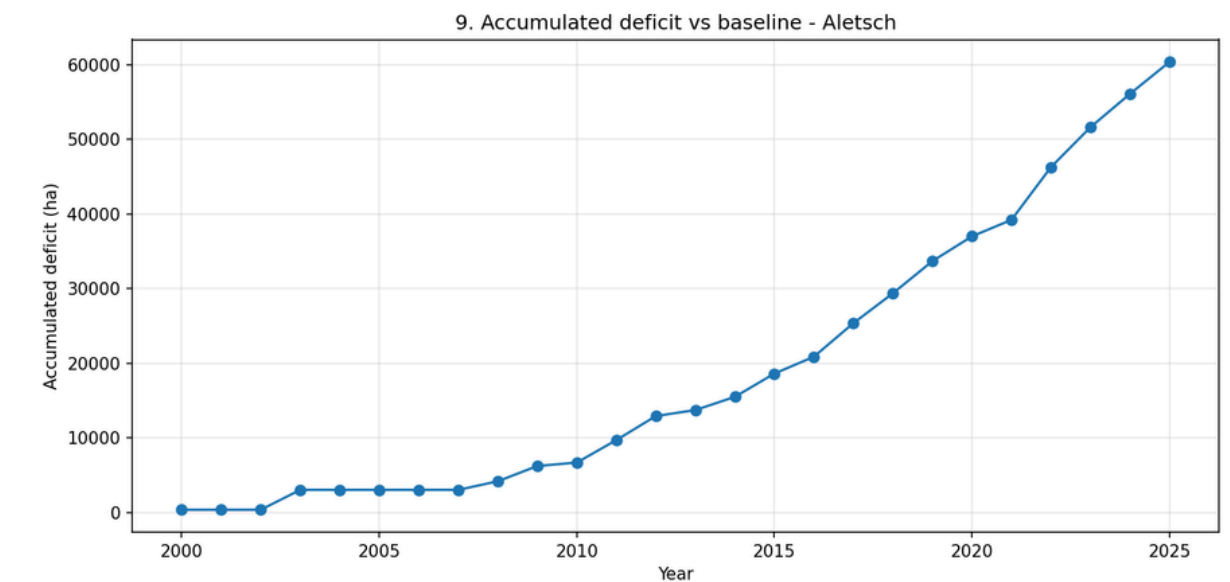
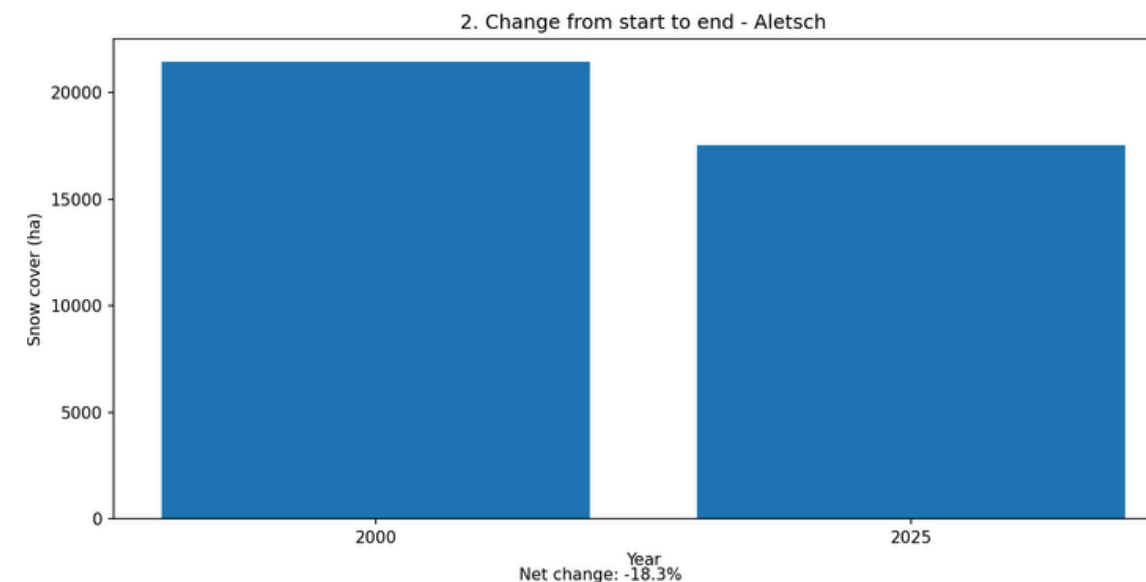
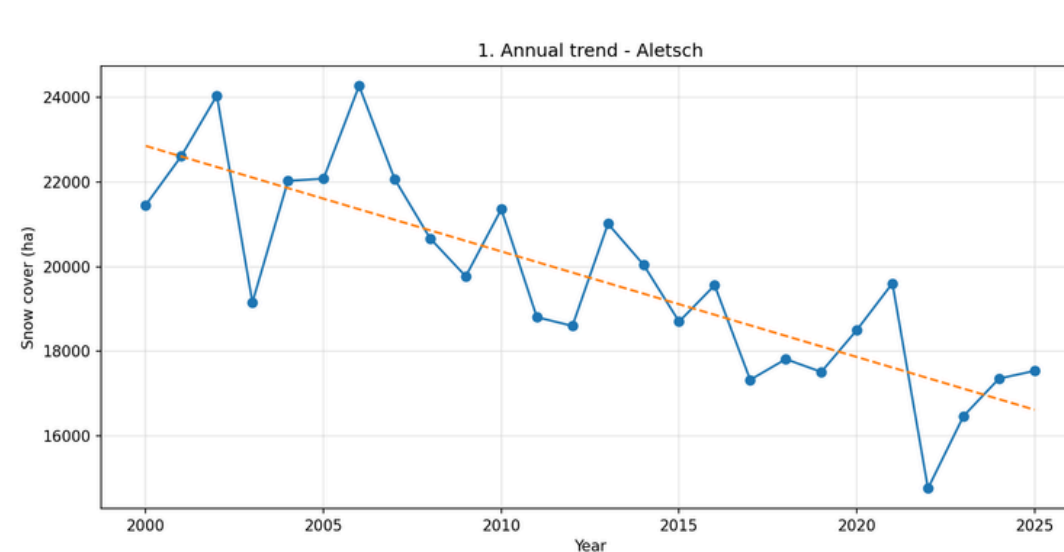
Cumulative snow-cover deficit relative to the study baseline in Tirol (Ötztal Alps), Austria.

The graph summarizes how annual snow-cover shortfalls accumulate through time, showing that recent deficits are not isolated anomalies but part of a persistent long-term decline.

Source: MODIS/Terra MOD10A1.061 annual snow-cover estimates derived in Google Earth Engine.



Data We Found - Snow Coverage - Switzerland



Annual snow-covered area in Aletsch, Switzerland, 2000–2025.

Derived from annual snow-cover estimates based on satellite observations processed in Google Earth Engine. The series shows interannual variability, but the long-term trajectory is downward, indicating a progressive reduction in snow-covered area across the study period.

Source line:

Source: satellite-derived annual snow-cover series processed in Google Earth Engine.

Change in snow-covered area between 2000 and 2025 in Aletsch, Switzerland.

This comparison summarizes the net difference between the first and last year of the series, showing an overall 18.3% decline in snow-covered area during the study period.

Source line:

Source: satellite-derived annual snow-cover series processed in Google Earth Engine.

Accumulated snow-cover deficit relative to the study baseline in Aletsch, Switzerland.

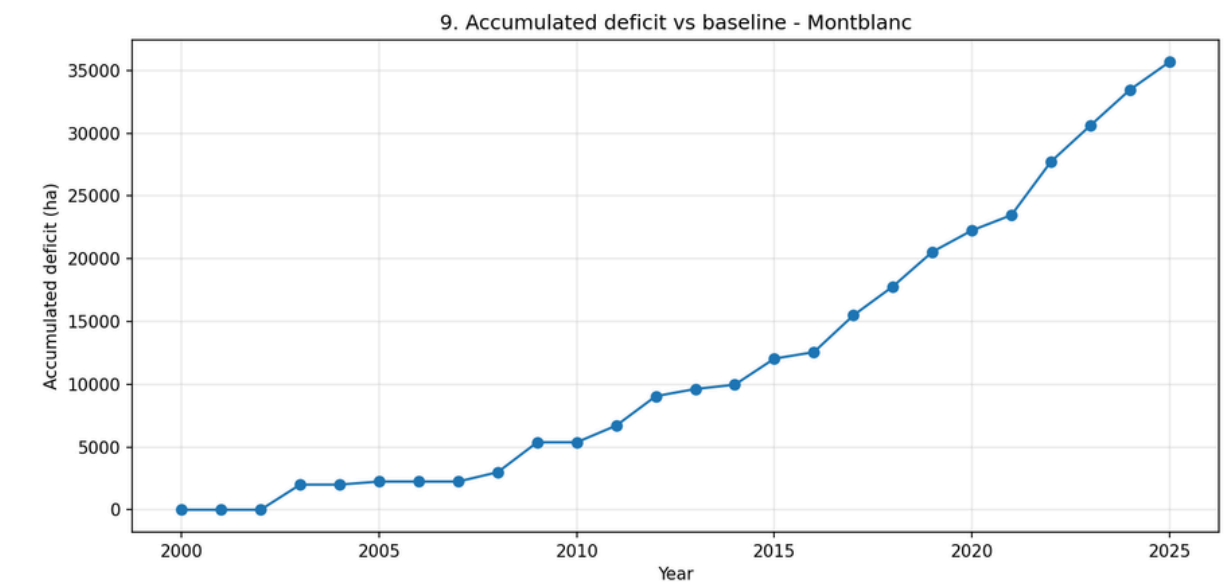
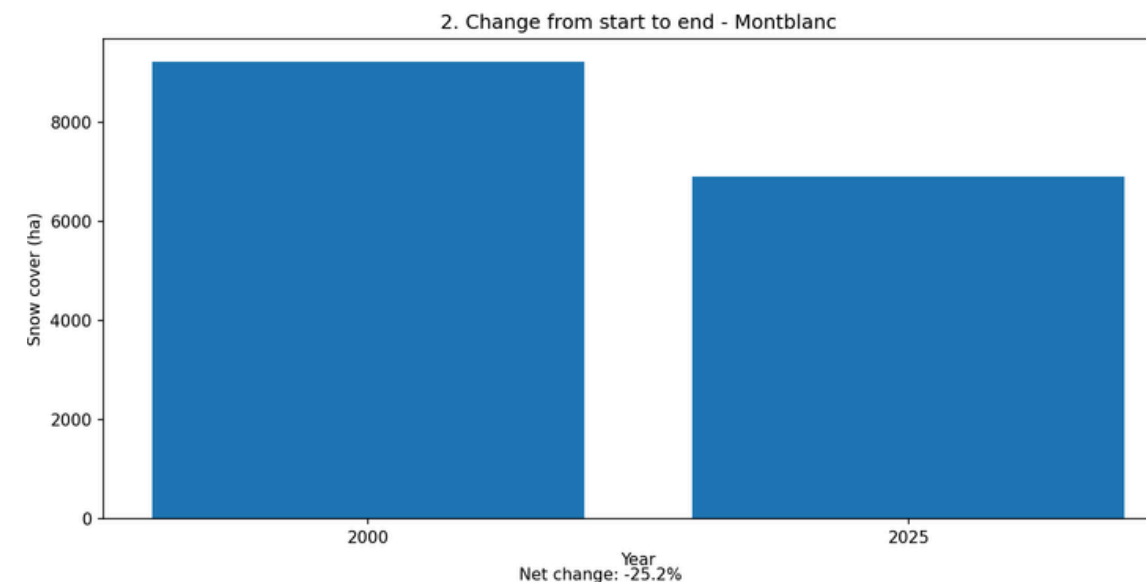
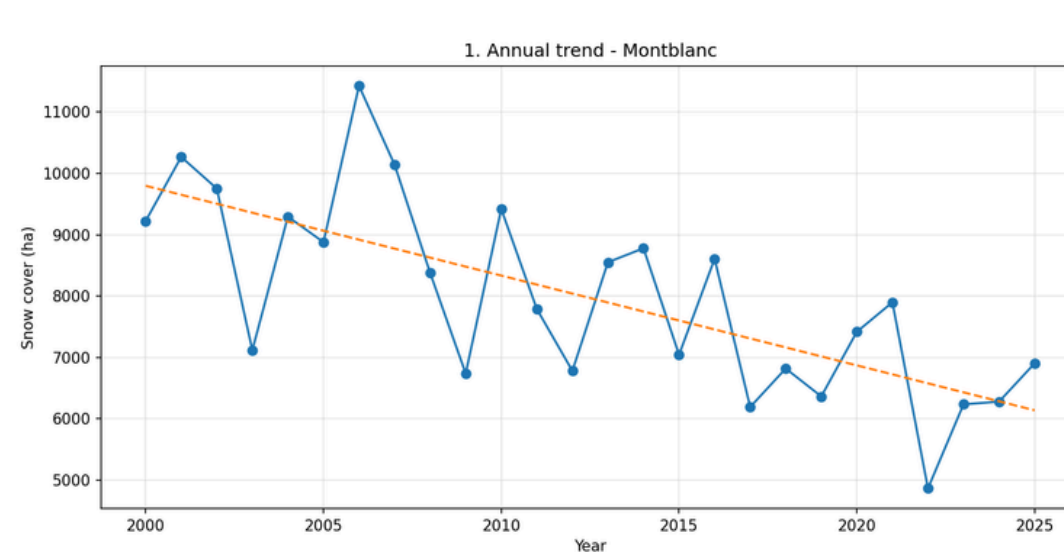
The cumulative deficit curve shows how annual shortfalls relative to the baseline build over time. Rather than reflecting a single anomalous year, the pattern indicates a persistent and compounding reduction in snow-covered area across the 21st century.

Source line:

Source: satellite-derived annual snow-cover series processed in Google Earth Engine.



Data We Found - Snow Coverage - France



Annual snow-covered area in Mont Blanc (Chamonix area), France, 2000–2025.

Derived from annual snow-cover estimates based on satellite observations processed in Google Earth Engine. The series shows marked interannual variability, but the long-term trajectory is downward, indicating a sustained reduction in snow-covered area across the study period.

Source: satellite-derived annual snow-cover series processed in Google Earth Engine.

Change in snow-covered area between 2000 and 2025 in Mont Blanc (Chamonix area), France.

This comparison summarizes the net difference between the first and last year of the series, showing an overall 25.2% decline in snow-covered area during the study period.

Source: satellite-derived annual snow-cover series processed in Google Earth Engine.

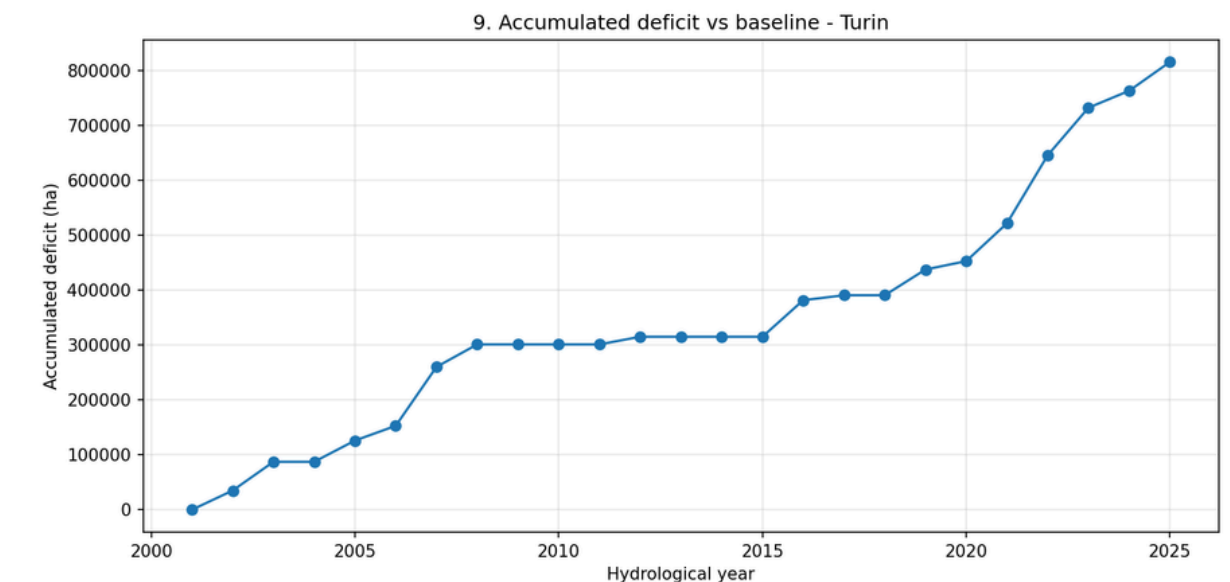
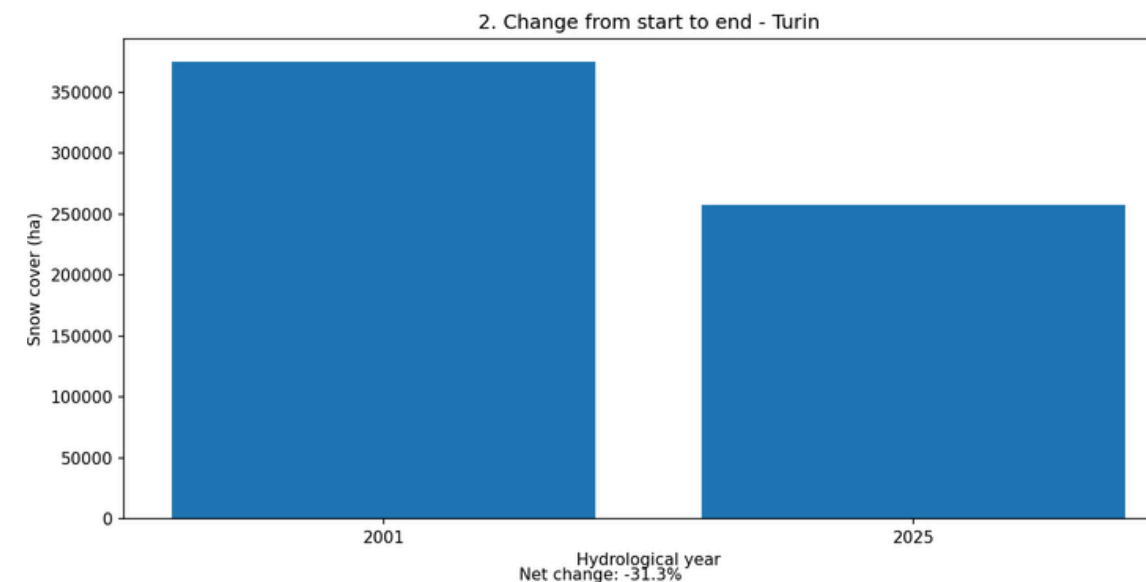
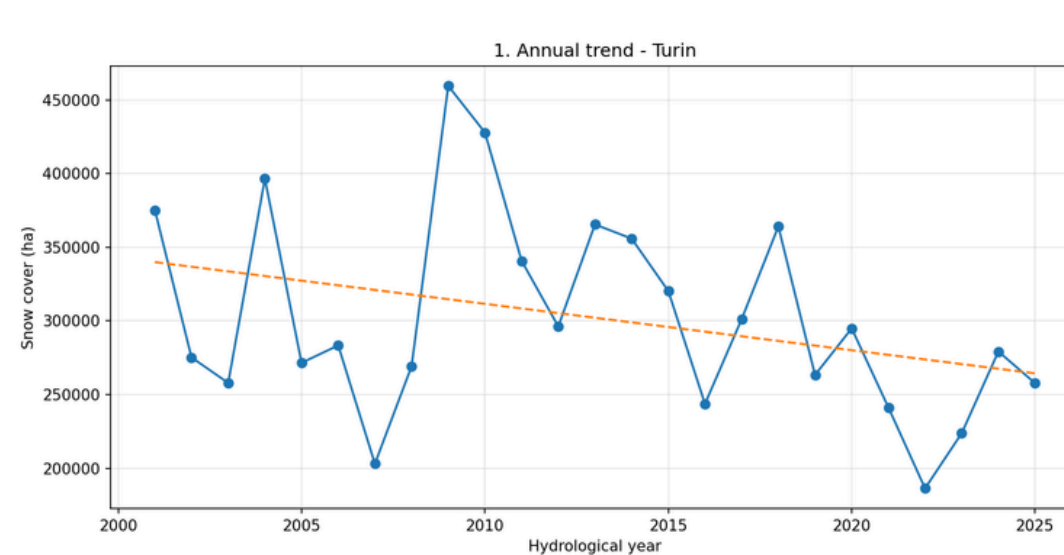
Cumulative snow-cover deficit relative to the study baseline in Mont Blanc (Chamonix area), France.

The graph summarizes how annual snow-cover shortfalls accumulate through time, showing that recent deficits are not isolated anomalies but part of a persistent long-term decline.

Source: satellite-derived annual snow-cover series processed in Google Earth Engine.



Data We Found - Snow Coverage - Italy



Annual snow-covered area in Turin, Italy, 2001–2025.

Derived from hydrological-year snow-cover estimates based on MODIS/Terra MOD10A1.061 Snow Cover Daily Global 500 m processed in Google Earth Engine. The series shows strong interannual variability, but the overall trajectory is downward, indicating a sustained reduction in snow-covered area across the study period.

Source: NASA NSIDC MODIS/Terra MOD10A1.061 (500 m, daily); processed in Google Earth Engine.

Change in snow-covered area between 2001 and 2025 in Turin, Italy.

Based on hydrological-year snow-cover estimates derived from MODIS/Terra MOD10A1.061. The comparison highlights the net percentage change in snow-covered area across the study period, showing an overall 31.3% decline between the first and last year of the series.

Source: NASA NSIDC MODIS/Terra MOD10A1.061 (500 m, daily); analysis by authors in Google Earth Engine.

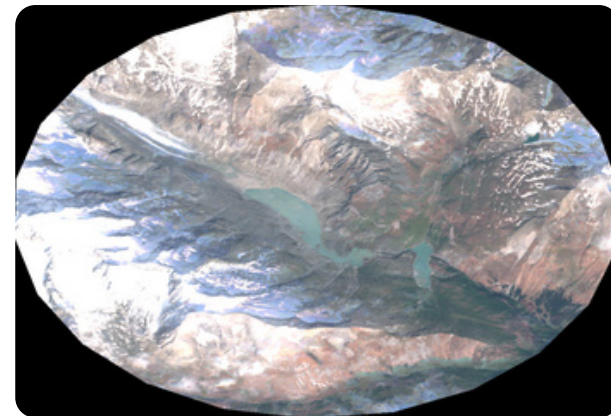
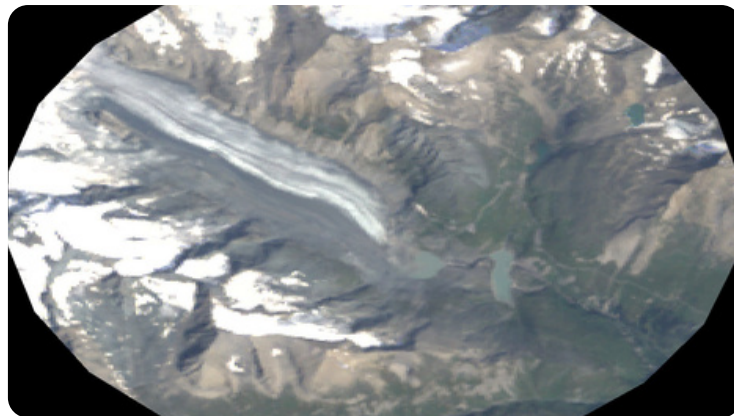
Cumulative snow-cover deficit relative to the study baseline in Turin, Italy.

The graph summarizes how annual snow-cover shortfalls accumulate through time, showing that recent deficits are not isolated anomalies but part of a persistent long-term decline across successive hydrological years.

Source: MODIS/Terra MOD10A1.061 hydrological-year snow-cover estimates derived in Google Earth Engine.



Data We Found - Lake Coverage

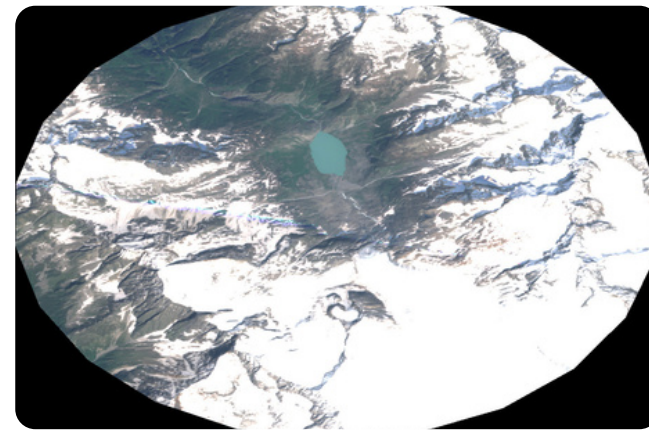


Lake Pasterze, Austria, 1990.

Landsat 5 TM (1990)

Derived from Landsat 5 Thematic Mapper (TM) imagery in Google Earth Engine. The image shows Lake Pasterze, Austria, on August 15, 1990, under late-summer conditions to reduce seasonal variability. It was used as the historical reference for evaluating glacier extent and proglacial lake conditions.

Source: USGS Landsat 5 Level 2, Collection 2, Tier 1 (TM); processed in Google Earth Engine.

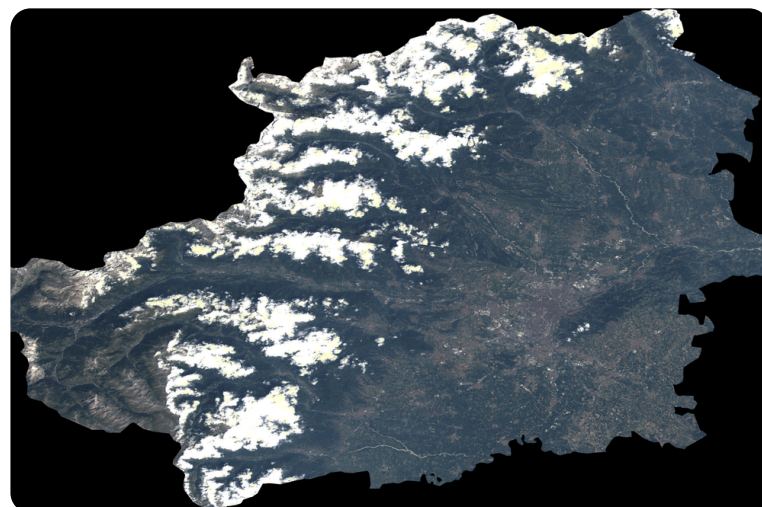


Lake Pasterze, Austria, 2025.

Copernicus Sentinel-2 MSI (2025)

Derived from Copernicus Sentinel-2 MultiSpectral Instrument (MSI) imagery in Google Earth Engine. The image shows Lake Pasterze, Austria, on August 15, 2025, under comparable late-summer conditions. It was used to assess recent landscape change, including glacier retreat and proglacial lake expansion.

Source: Copernicus Sentinel-2 MSI Level-2A (SR, Harmonized); processed in Google Earth Engine.



Turin, Italy, 2000

Landsat 5 TM (2000)

Derived from Landsat 5 Thematic Mapper (TM) imagery in Google Earth Engine. The image shows the Turin metropolitan area and surrounding Alpine foothills, Italy, using a late-summer composite (July–September 2000) to reduce seasonal variation and cloud effects. It was used as the historical baseline for assessing snow presence in the surrounding mountains and broader land-surface conditions across the study area.

Source: USGS Landsat 5 Collection 2 TOA imagery (TM), processed in Google Earth Engine.

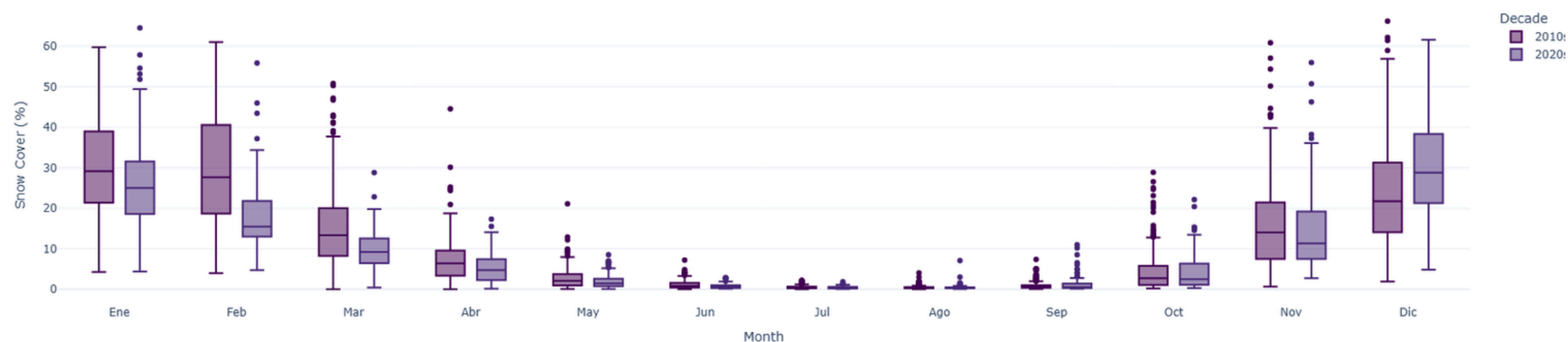
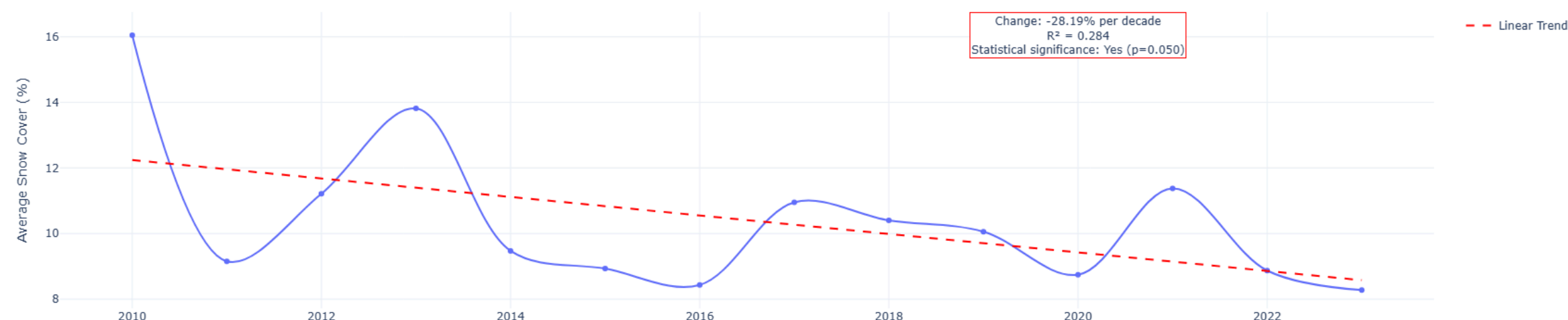


Alps Data

The 25-year trajectory studied through Python is definitively downward since the 2020s consistently show lower medians and tighter distributions across nearly every month of the year.

MOD10A1.061 Terra Snow Cover Daily Global 500m

Snow Cover Annual Trend

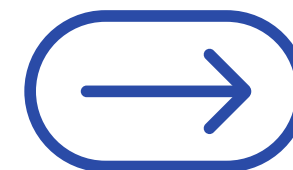
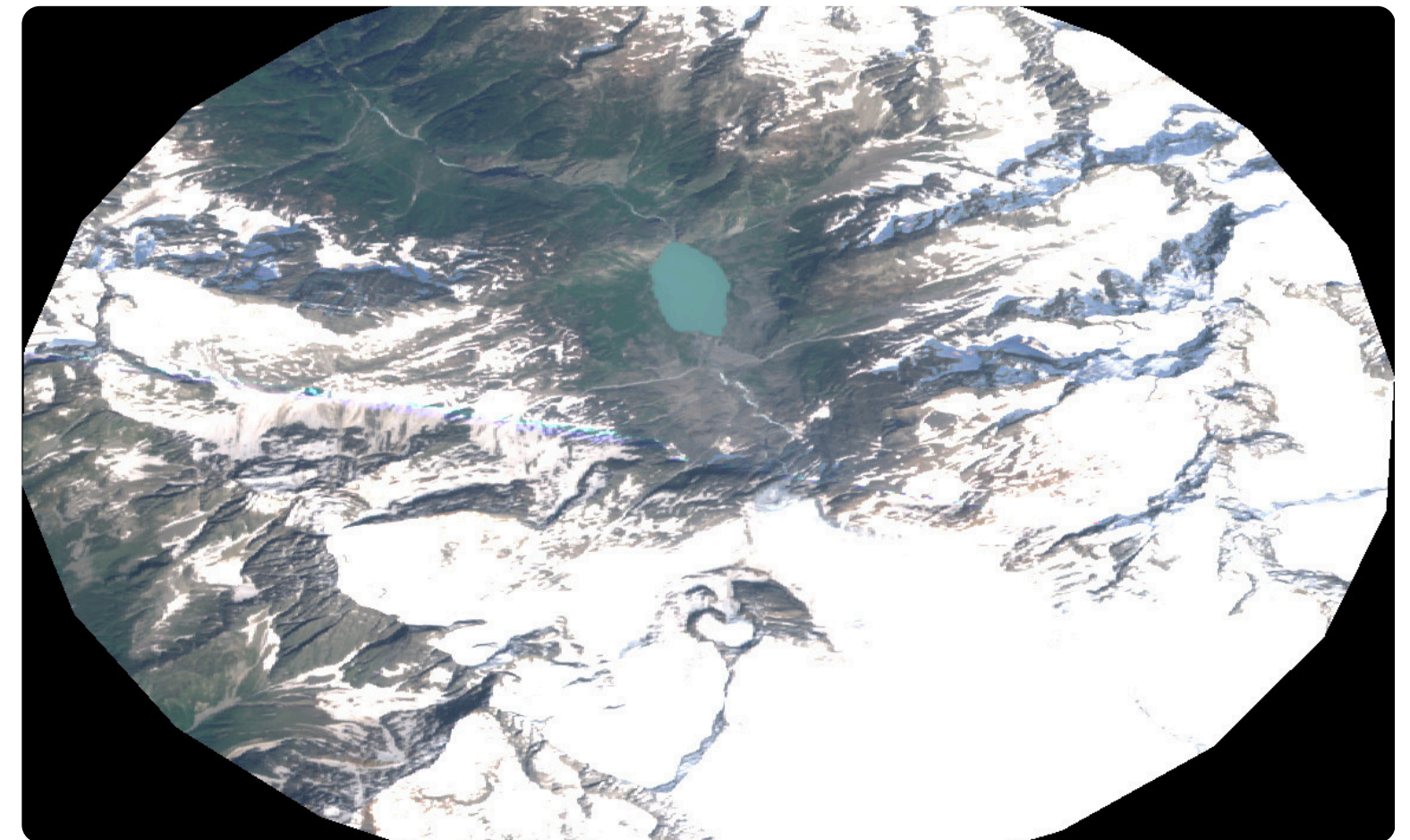


Monthly Snow Distribution Per Decade



Key Findings

Visual comparisons of Lake Pasterze (Austria) and Lake Triftsee (Switzerland) between 1990 and 2025 provide tangible proof of glacier retreat and proglacial lake expansion. The cumulative deficit curves confirm these are not isolated anomalies but part of a persistent, compounding long-term decline throughout the 21st century.



These findings directly address our track's problem statement by providing objective, satellite-verified evidence that can debunk false claims minimizing or denying Alpine glacier retreat.



Data Sources

- ➔ NASA NSIDC MODIS/Terra MOD10A1.061 Snow Cover Daily (2000-2025)
- ➔ Copernicus Sentinel-2 (ESA)
- ➔ GLAMOS Switzerland
- ➔ Google Earth Engine

Peak News

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OPTICAL SATELLITE · 10M

[sentinel.esa.int](#)

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MULTISPECTRAL SATELLITE · 30M

[landsat.usgs.gov](#)

95%

3 GLAMOS Switzerland

GLACIER MONITORING NETWORK

[glamos.ch](#)

99%

4 MeteoSwiss

METEOROLOGICAL DATA

[meteoswiss.admin.ch](#)

96%

5 Semantic Scholar

SCIENTIFIC LITERATURE

[semanticscholar.org](#)

88%

6 IPCC AR6

PEER-REVIEWED SCIENTIFIC REPORT

[ipcc.ch](#)

100%

SCIENTIFIC REFERENCE REPOSITORIES

Nature Climate Change

Nature

IPCC Sixth Assessment Report

IPCC

World Meteorological Organization

WMO

Copernicus C3S

Copernicus

Science / AAAS

Science



Impact



We believe Earth Observation Data is essential to shift climate discourse from opinion to objective reality.

By transforming datasets like MODIS and Sentinel-2 into clear visual evidence, we empower journalists to dismantle false narratives with ground truth, ensuring public debate is anchored in physical facts rather than political rhetoric.



PeakNews supports [UN SDG 13 \(Climate Action\)](#) by providing clear, indisputable visualizations of glacial retreat and shrinking snow windows.



The true value of satellite data is its ability to provide temporal and spatial context invisible from the ground. It replaces "he-said, she-said" reporting with objective proof.

When headlines claim a glacier is growing, PEAKNEWS satellite imagery can instantly reveal decades of decline. This transparency is the ultimate antidote to the "Climate Scam" narratives that thrive in data vacuums.

So What?

